How Nature Builds 10,000 Different Fishes

The Visual Guide for Everyone (13-Year-Old Level with Diagrams)

Why a 109-Year-Old "Rubber Graph-Paper" Idea Was Right — And Why Computers Missed It

Infinite Morphospace Project · `https://github.com/fractastical/morpholib`

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The 60-Second Story in One Sentence

Global growth rules (D'Arcy Thompson's curved rubber sheet) shape the *outside* of the fish for swimming over 150 million years, while competition with neighbors rewires the *jaws and snout* on the inside (3× to 9× faster) so sister species can share the same coral reef without fighting over the same food.

Step 1: D'Arcy Thompson's 1917 "Rubber Graph-Paper" Idea

In 1917, a Scottish scientist named **D'Arcy Thompson** asked a simple question: *When one fish species evolves into another, does nature have to redesign every single bone from scratch?*

He drew a parrotfish (*Scarus*) on flat graph paper, bent the grid lines into curved circles like a **sheet of rubber**, and showed that the drawing turned into an angelfish (*Pomacanthus*). He argued that an embryo only needs to turn a few global **"growth knobs"** (stretching, slanting, or curving the whole body at once) to transform one body plan into another.

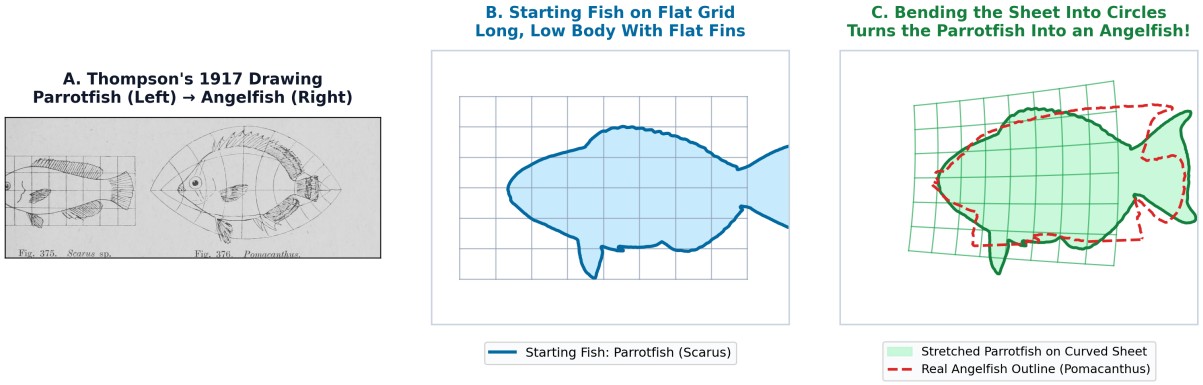


Figure 1: **Step 1 — D'Arcy Thompson's 1917 Rubber Graph-Paper Idea.** **(A)** Thompson's original 1917 drawing (p. 749). **(B)** A parrotfish on flat graph paper. **(C)** Bending the graph paper into coaxial circles transforms the parrotfish silhouette into an angelfish.

Step 2: Mistake #1 — Why Computers Thought Thompson Was Wrong ("The 17-Dot Head Trap")

For the last 30 years, biologists tested Thompson's idea on computers and concluded it didn't work across different fish families. **Why did their computer tests fail?**

Because instead of tracing the **whole outline of the fish**, standard fish databases only measure **17 dots on the skeleton**:

- **9 of those 17 dots** (53% **of the math weight) are crammed inside the head and jaws** (the yellow circles in Panel A below).
- The other **8 dots** (47%) only mark where the bases of the fins touch the body wall (green squares).
- **Zero dots** (0%) **are on the actual outer fins or tail!**

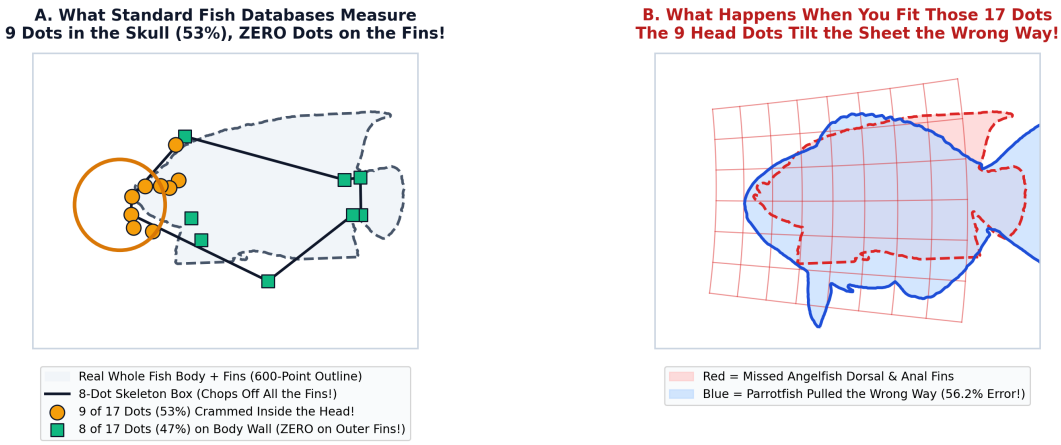


Figure 2: **Step 2 — The "17-Dot Head Seesaw Trap." (A)** Standard fish databases put 9/17 dots (53%) inside the skull and 0 dots on the outer fins. **(B)** When you fit a rubber sheet to those 17 dots, the 9 head dots act like a heavy seesaw and tilt the back of the sheet the wrong way (56.2% mismatch — worse than not stretching the fish at all!).

Why the 17 Dots Failed

Imagine trying to tailor a full-body wetsuit by measuring **9 points on someone's face** and **0 points on their arms or legs**. Because heads and fins evolve differently across fish families, forcing the rubber sheet to fit the 9 head dots swings the back of the sheet in the wrong direction!

Step 3: Mistake #2 — Why "Just Stretching a Box" Fails True Anatomy, and Why Thompson's Curves Win

When we switch from 17 skeleton dots to a **600-point outline of the whole fish**, we have to be careful of a second trap: *How do we know Thompson's slanted and curved grids are really*

needed, instead of just stretching a plain rectangle taller or wider ("Plain Box Stretch")?

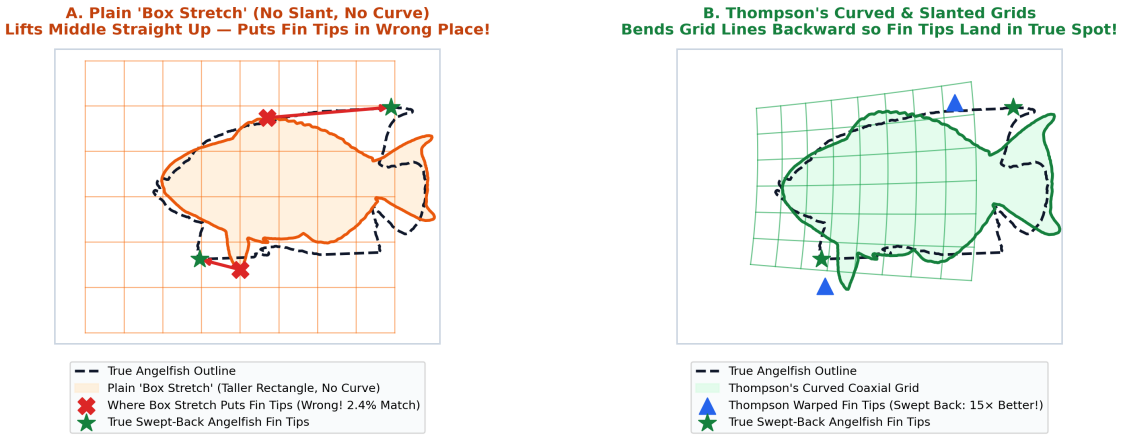


Figure 3: **Step 3 — Plain "Box Stretch" vs. Thompson's Curved Grids. (A)** A plain rectangular Box Stretch (no slant, no curve) lifts the middle of the back straight up (red **X**), missing the swept-back fin tips of the angelfish (2.4% accuracy). **(B)** Thompson's curved coaxial grid bends the grid lines backward so the fin tips land in their true swept-back position (15× more accurate!).

- **The "Shadow Cheat":** If you only ask a computer to overlap two **unlabeled black shadows**, a plain rectangular Box Stretch can "cheat": on a pufferfish → ocean sunfish, it stretches the *middle of the pufferfish's belly* straight up and down to cover the *back fins* of the sunfish! And on a parrotfish → angelfish (Panel A above), a Box Stretch lifts the *middle of the back* straight up (red **X**), missing the swept-back fin tips completely (**only** 2.4% **match!**).
- **Why Thompson's Curved Grids Win:** Once you tell the computer which part of the outline is the **Dorsal Fin Tip**, which part is the **Anal Fin Tip**, and where the fins attach along the body (without the 9 internal skull dots), a plain Box Stretch fails, and **Thompson's slanted and curved grids jump by** +13% **to** +48% **accuracy** across all six 1917 pairs!

Step 4: The Big Picture Across 10,303 Species — Two Simple Rules of Fish Evolution

To be sure our answer wasn't a fluke of just a few fish, we tested **three independent datasets** totaling **10,303 fish species across 987 genera and 440 families**. We found that fish evolution follows **two simple rules at the same time**:

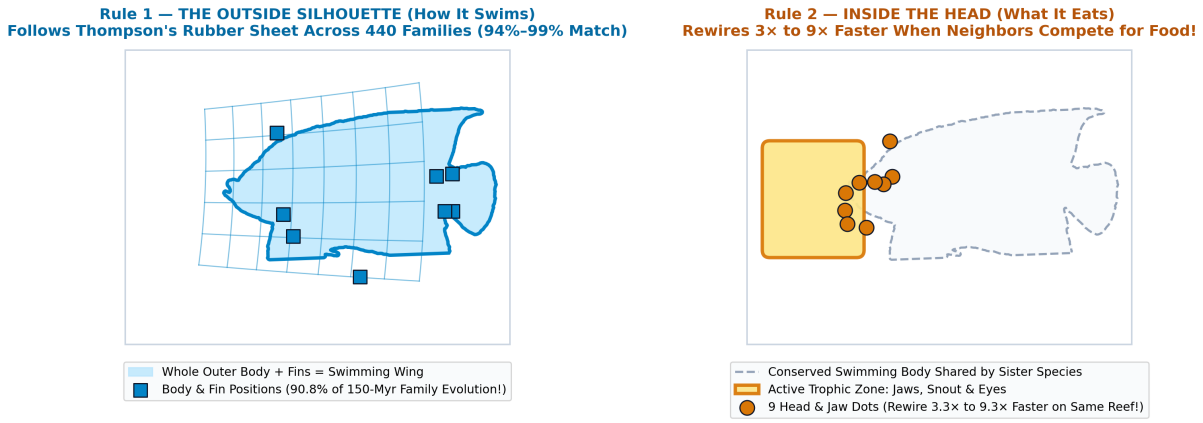


Figure 4: **Step 4 — The Two Rules of Fish Evolution Across 10,303 Species. Left (Rule 1 — The Outside Silhouette):** The whole outer body and fins work together like a swimming wing and follow D'Arcy Thompson's curved rubber sheet across 440 families (94%–99% alignment). **Right (Rule 2 — Inside the Head):** When sister species live side-by-side on the same coral reef, their jaws, snouts, and eyes rewire 3.3× to 9.3× faster than their bodies so they can eat different foods without competing.

Rule 1 — The Outside Silhouette (How the Fish Swims)

Water doesn't flow around an invisible internal skeleton; it flows around the **whole outer silhouette** of the body and fins acting like an airplane wing. Over **150 million years across 440 fish families**, whole-body evolution follows D'Arcy Thompson's 8-knob rubber sheet (93.5% to 98.9% alignment!). Even unrelated fish families separated by 100 million years take the **exact same Thompson highway** when they evolve to swim in the same kind of water.

Rule 2 — Inside the Head (What the Fish Eats)

Why did the 9 head dots break the global rubber sheet? Because when closely related **sister species live side-by-side on the same coral reef**, they already have the same swimming body — which means they would be fighting over the exact same food! To avoid competing, sister species rapidly rewire their **jaws, snouts, and eyes** 3.3× **to** 9.3× **faster than their bodies** (85.4% of within-genus evolution): one cousin evolves jaws to crunch coral, another sucks in tiny plankton, and another scrapes algae!

Step 5: The Final Scoreboard in Four Simple Numbers

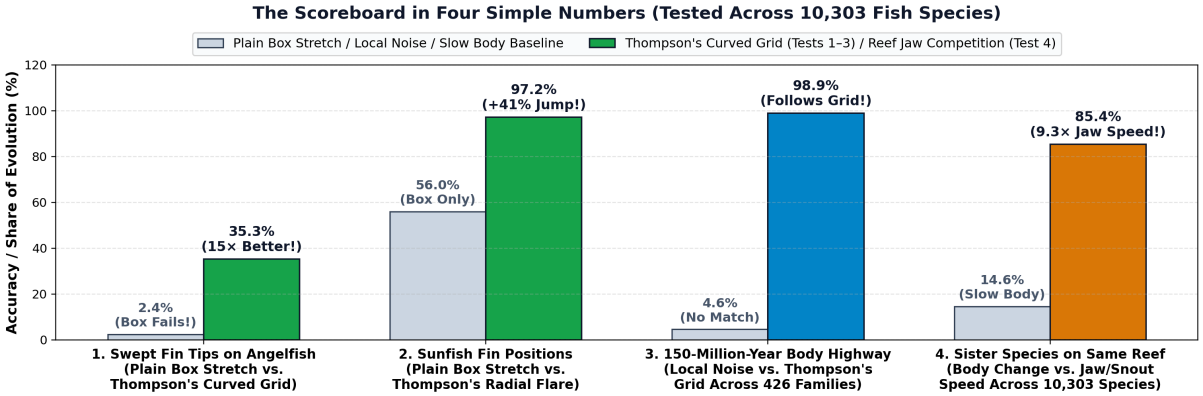


Figure 5: **Step 5 — The Four-Number Scoreboard Across 6 Classic Pairs & 10,303 Species.** Comparing plain Box Stretch / baseline assumptions (grey bars) against Thompson's curved grids (Tests 1–3) and reef jaw competition (Test 4).

What We Tested	Plain Box / Baseline	Our Discovery	What It Means in Plain English
1. Angelfish Swept Fin Tips	2.4% (Box Fails)	**35.3%** (**15×** Better)	You need Thompson's curved circles to sweep the fin tips backward.
2. Ocean Sunfish Fin Bases	56.0% (Box Only)	**97.2%** (**+41%** Jump)	You need Thompson's exponential flare to match the giant tailfins.
3. 150-Myr Body Highway	4.6% (Local Noise)	**98.9%** (Follows Grid!)	Across 426 families, whole body plans evolve along Thompson's grid.
4. Reef Sister-Species Jaws	14.6% (Slow Body)	**85.4%** (**9.3×** Jaw Speed!)	Across 10,303 species, neighbors on the same reef rapidly rewire their jaws to share food.